TECHNICAL PROPOSAL

VoiceKey

Offline real-time English-Vietnamese translation for short, face-to-face conversations

Project	VoiceKey
Submission type	Phase 2 technical proposal
Version	v2.1
Date	21 August 2026
First platform	Android

Technical concept and validation plan

1. Executive Summary

VoiceKey is a proposed offline English-Vietnamese translation system for short, face-to-face conversations. It helps Vietnamese frontline workers communicate with English-speaking trainers, supervisors, technicians, or visitors without waiting for an interpreter or relying on stable internet access. The first use case is a two-person conversation in an indoor setting such as onboarding, maintenance, warehouse work, or field service. VoiceKey is a communication aid, not a safety system. Critical instructions still need local procedures and human read-back.

The first prototype is an Android app. It runs speech recognition, translation, and optional text-to-speech (TTS) locally on the phone. It shows partial text while someone speaks and a large bilingual result after a natural pause. Conversation is automatic and turn-based. VoiceKey v1 does not interpret two people speaking at the same time.

The team will also test a small USB-C companion with two close microphones, an optional button, and a recording LED. The phone provides the screen, battery, storage, speaker, and AI compute. A USB drive that only stores models does not improve translation speed or conversation quality by itself. The companion stays only if it improves audio capture, turn control, or user trust over the phone-only baseline. A full AI compute module is a later research track that needs its own processor, memory, power, and thermal design.

The central question is simple: can VoiceKey provide useful offline English-Vietnamese translation on Android? It compares phone-only and companion-assisted use under the same conditions, then keeps the hardware only if the measurements support it.

1.1 Problem Overview

Short spoken instructions are difficult when people do not share a language. A generic app can help, but it may depend on network access, phone-specific microphone behavior, and awkward hand-held interaction. The problem becomes more serious when people need to confirm a number, a task, a part, or a sequence quickly.

1.2 Proposed Solution

VoiceKey is an Android-first, offline, turn-based English-Vietnamese translation app. An optional USB-C companion can add close microphones, a visible recording state, and a physical control; the Android phone runs the local models and shows the bilingual text.

1.3 Key Value Proposition

1. **Offline runtime.** After a signed language pack is installed, speech recognition, translation, and optional TTS must run with the network disabled.
2. **A shared bilingual screen.** Both people see the source text and translation before acting.
3. **Automatic, turn-based conversation.** Partial text appears during speech; final translation appears after a natural pause. A button remains an optional fallback in noisy settings.
4. **A hardware decision based on evidence.** The USB-C companion remains in scope only if it improves the same conversation task over phone-only use.

2. Problem Definition and Target Users

2.1 Problem Statement

The first target situation is a short face-to-face conversation between a Vietnamese worker and an English-speaking colleague. The user needs quick bilingual text, not a long meeting transcript or a general business assistant. They may be in a noisy indoor setting, may not have reliable internet, and may need to check names, numbers, instructions, or technical terms. Existing translator products show useful interaction patterns, but VoiceKey still has to prove that it helps in this setting.

2.2 Impact Analysis

Impact area	Current difficulty	Potential consequence	VoiceKey response to test
Understanding	Spoken instructions can be missed or misheard	Repetition and unclear confirmation	Show source and translated text together; ask for a repeat when uncertain
Productivity	People stop to type, call a bilingual colleague, or repeat a phrase	Slower training and task handoff	Short spoken turns and large shared text
Connectivity	Wi-Fi or mobile service can be weak, costly, or prohibited	Cloud-dependent translation may stop	Local runtime after pack installation
Privacy	Audio or text can be sent to an external service without a clear policy	Lower user trust and policy concern	No silent cloud fallback; local history is opt-in
Accessibility	A spoken output alone is hard to verify	One person may not know what the system heard	Visible bilingual transcript and optional spoken playback

2.3 Target Users and Use Cases

User segment	Language need	Primary context	First use case	Priority
Vietnamese frontline worker	Vietnamese <-> English	Factory, warehouse, field service	Receive and confirm a short instruction	High

User segment	Language need	Primary context	First use case	Priority
English-speaking trainer or supervisor	English <-> Vietnamese	Onboarding, quality check, maintenance	Explain a process and confirm understanding	High
Field technician or contractor	English <-> Vietnamese	Installation, inspection, visitor support	Clarify a fault, part, measurement, or appointment	Medium
Operations facilitator	English <-> Vietnamese	Guided two-person discussion	Review a bilingual transcript with consent	Medium

2.4 Key Design Constraints

Constraint	Prototype target or rule
Internet dependency	No required network connection at runtime after the signed pack is installed and verified
Conversation mode	Automatic turn detection; partial text during speech; final translation after a pause
Latency	P50 <= 2.0 seconds and P95 <= 3.5 seconds from speech endpoint to final translated text for a 2 to 6 second turn
First language pair	English <-> Vietnamese text, speech recognition, and optional local TTS
Environment	Indoor pilot; test 70 to 85 dBA noise. Outdoor and very loud use are out of scope for v1.
Safety boundary	Never present translation as an instruction to act without normal local confirmation procedures
Privacy boundary	Audio is not retained by default. Conversation history is opt-in, encrypted on the phone, and deletable.
Platforms	Android first. Evaluate iOS as phone-only; do not promise a custom USB-C iPhone path.

3. Business Solution and Innovation

3.1 Industry Problem and Solution Fit

VoiceKey focuses on short, face-to-face conversations where visible bilingual text, offline behavior, and clear device state matter more than broad language coverage or travel features.

Existing approach	Useful feature	Trade-off for the VoiceKey use case	VoiceKey design response
Phone-only translator app	Low cost, familiar screen	Microphone position and interaction vary by phone; no physical recording state	Establish the phone-only baseline, then test whether the companion helps
Translation earbuds	Wearable conversation format	Bluetooth pairing, battery, and phone-app interaction add steps; text can be secondary	Use a wired companion only if it improves capture and keep the phone transcript primary

Existing approach	Useful feature	Trade-off for the VoiceKey use case	VoiceKey design response
Standalone translator	Dedicated device and screen	Adds another battery, display, radio, charging path, and compute platform	Reuse the phone screen and local compute in v1
Cloud translation service	Can use large remote models	Needs a network and moves conversation data beyond the phone	Require a local offline pipeline after pack installation

The Timekettle W4 Pro tutorial shows a phone-coupled earbud workflow, while Pocketalk's getting-started tutorial shows a dedicated handheld workflow. These videos demonstrate workflows, not whether VoiceKey performs better. VoiceKey instead tests phone-visible bilingual text and an offline Android runtime. Direct demos are listed in Appendix B.

3.2 Innovation and Competitive Strengths

Dimension	VoiceKey v1 decision	What must still be proven
Offline operation	Local language pack and no silent cloud fallback	Airplane Mode test, network-egress check, and pack recovery behavior
Real-time interaction	Partial text while speaking; final translation after a pause	Endpointing, response time, and user acceptance
Shared understanding	Large bilingual transcript keeps both source and translation visible	Users can read it quickly in the target environment
Audio and control	Optional close microphones, button, and LED	Measurable benefit over phone microphone and touch-only control
Privacy	Local processing, no audio saved by default	Secure pack handling, history controls, and pilot consent flow
Hardware scope	No screen, battery, modem, or AI computer in v1	USB power, reconnect, heat, and route stability on supported phones

The USB companion is not assumed to improve translation on its own. The phone handles AI and the user interface; the companion is tested only for tasks hardware can improve. This avoids paying for a second computer before it proves its value.

3.3 Cost Structure and Pilot Boundary

VoiceKey has two cost layers. The companion is a low-power audio accessory. The app, offline model integration, testing, and support are the larger early investment. Public component prices and labor data are for planning only, not supplier quotes or selling prices.

Cost layer	Planning band	What it covers
EVT companion, 10 units	USD 45 to 70 per unit	Board, simple enclosure or pigtail, basic assembly, and test

Cost layer	Planning band	What it covers
Pilot companion, 30 to 40 units	USD 38 to 60 per unit	Lower assembly cost, with spares and rework still material
100-unit preproduction companion	USD 30 to 45 per unit	Better component and assembly pricing; still not a production quote
10-user BYOD pilot	USD 2.835k to 9.210k direct cash	Dongles, simple fixture, packaging, field support, and replacement reserve; excludes NRE
30-user BYOD pilot	USD 5.582k to 17.640k direct cash	Same categories at the larger pilot size; excludes NRE
Lean MVP engineering	USD 65k to 95k	Hardware, Android, local inference, QA, pilot preparation, and pre-scan advisory
More complete first-release engineering	USD 95k to 145k	Full workstream planning band; formal certification remains quote-required

The pilot should measure repeated explanations, time until both people confirm understanding, support requests, and user acceptance. Do not estimate return on investment until those values are measured. Before procurement, the team needs a selected schematic, two comparable manufacturing quotes, a warranty and replacement plan, and a clear policy for customer-provided versus loaner phones. Section 5.3 gives component and cost anchors. The cost model lists direct pilot cash: dongles, fixtures, labels or shipping, replacement reserve, and field support. It excludes NRE, loaner phones, formal certification, and lab capital unless a row explicitly says otherwise.

3.4 Commercial Model

Start with a small business pilot, not a mass-market travel gadget. The pilot kit includes the companion only if it passes the baseline comparison, plus the Android app, an English-Vietnamese pack, setup guidance, and a test report. A later commercial offer could separate hardware, site onboarding, and an optional support service for signed pack updates and glossary governance. The offline translation core should not depend on a recurring cloud-inference fee.

This proposal does not set a retail price. Comparator retail prices are for context only because they bundle different phones, connectivity, warranty, language packs, support, tax, and distribution terms. The base case assumes a customer-provided Android phone from a supported-device list.

4. AI Approach and Technical Design

4.1 System Pipeline Overview

VoiceKey uses a cascade so each stage can be inspected and replaced independently. This is a sensible starting point for English-Vietnamese: the PhoST benchmark reports an English-Vietnamese cascade result that outperformed its end-to-end comparison. This supports the architecture choice, not VoiceKey performance.

Real-time translation flow

Phone or companion microphone

v

Speech detection and partial speech text

v

Final speech text after a natural pause

v

Local English-Vietnamese translation

v

Large bilingual transcript

v

Optional local TTS

The signed local language pack supplies speech, translation, and optional TTS resources.

Normal use does not require Push-to-Talk. The system detects speech and the end of a turn. The button is an optional mute or fallback control in noise. When the phone plays optional text-to-speech, live capture pauses in v1 so the speaker output is not recognized as a new user turn. Full-duplex translation, overlap handling, and acoustic echo cancellation are outside the first prototype.

4.2 Module-by-Module Design

The following are engineering budgets and candidate classes, not measured VoiceKey results. They do not select a final model or voice.

Module	Candidate class	Size budget	Latency budget	Purpose and selection rule
Speech detection	Local VAD, such as Silero VAD	About 2 MB class	<20 ms per chunk target	Detect start and end of a turn; verify behavior in noisy slices
Speech recognition	Benchmark Whisper base and small variants through whisper.cpp, then compare them with PhoWhisper variants on named Vietnamese ASR test sets	Measured per candidate	Partial update <=500 ms after a one-second audio window; final text <=1.0 s after endpoint	Record actual quantized file size, runtime memory, accuracy, heat, and latency. Do not assume one shared size band.
Translation	OPUS-MT EN-VI and OPUS-MT VI-EN evaluation candidates, with release-license review	80 to 350 MB budget	<=1.5 s target after final text	Compare bilingual adequacy, protected terms, and device performance before choosing a shipping candidate
Text-to-speech	Local runtime such as sherpa-onnx with a separately reviewed voice	30 to 120 MB budget	<=0.7 s target before playback	Optional output only; do not ship a voice until license and quality review pass
Glossary and confidence rules	Local approved terms, number checks, and uncertainty flags	Small	Immediate	Protect names, part codes, numbers, and site terms

The Whisper model card says the released models are not real-time out of the box. Treat every model, quantization level, runtime, phone, and language pack as a separate benchmark candidate. PhoMT and PhoWhisper guide Vietnamese evaluation; they do not grant automatic commercial rights to model weights, voices, or runtimes.

4.3 On-Device Optimization and Memory Management

v1 keeps one pipeline in memory at a time. It streams short audio chunks, keeps partial text small, loads only the selected English-Vietnamese pack, and releases TTS resources when they are not needed. The initial engineering target is 1.5 GB of peak AI memory on the reference phone. Record pack size, startup time, temperature, and battery use for every candidate stack.

Start with a float baseline, then test INT8 candidates where the mobile runtime supports them. Test a weight-only, low-bit variant only if it reduces memory use or heat without a meaningful quality loss. A quantized candidate can enter the pilot only if bilingual adequacy is no more than two percentage points below its float baseline, it passes the 30-minute thermal test, and it meets the response-time target. These are prototype gates, not results.

Start with a CPU implementation. Add GPU or vendor acceleration only if the same model, pack, and phone pass accuracy, latency, heat, and stability tests. The USB companion has no direct access to the phone NPU, so it is not a shortcut to hardware acceleration.

4.4 Robustness and Edge Cases

Situation	Required behavior
Background noise	Show the original text, flag low confidence, and ask for a repeat rather than inventing a translation
Names, numbers, and part codes	Preserve protected terms and highlight uncertain segments
Fast speech	Ask the speaker to shorten or repeat the turn
Mixed English and Vietnamese	Preserve source text and request a clearer turn if the language route is unclear
TTS playback	Pause capture during playback in v1 to prevent echo loops
Missing or invalid pack	Show a clear local error. Do not use a hidden cloud fallback.
USB companion missing	Continue with the phone microphone and tell the user which input is active
Heat, power, or memory limit	Stop optional features with a visible status. Never create a result from incomplete processing.

4.5 Evaluation Plan

The test set combines public research data with consented short conversations from the intended setting. It covers quiet speech, 70 to 85 dBA noise, Northern, Central, and Southern Vietnamese accents, names, numbers, technical terms, and code-switching. Report speech accuracy, bilingual review of translation adequacy, P50/P95 response time, task completion, and failure rate. Use public slices such as PhoMT as one part of the evaluation. Do not publish an accuracy claim until the target-context test passes.

Prototype acceptance target	Pass condition before pilot
End-to-end response time	P50 <=2.0 seconds and P95 <=3.5 seconds from endpoint to final translated text on the reference phone
Protected terms and numbers	At least 90 percent correct on the held-out target-context set; each critical error is reviewed separately
Bilingual translation adequacy	At least 80 percent of final turns rated adequate or better by bilingual reviewers using a fixed rubric
Short-task completion	At least 80 percent of scripted tasks completed after normal human confirmation
Offline and stability	All required turns pass in Airplane Mode and in a network-enabled test that records zero VoiceKey outbound connections after the local pack is installed and verified. For the 30-minute session, report the pack version and hash, capture method, crashes, and route losses.

5. Hardware and Device Concept

5.1 Platform Selection and Justification

v1 puts the local AI on the phone and uses an optional low-power USB audio companion. It lets the team test the translation experience without assuming that a small USB peripheral can contain a full AI computer.

Criterion	Phone-only baseline	USB audio and interaction companion	VoiceKey Edge compute module	v1 decision
AI compute	Phone CPU, GPU, and supported mobile runtime	Still runs on phone	Own processor, AI accelerator, RAM, storage, and firmware	Phone hosts AI
Audio interaction	Built-in phone microphone and touchscreen	Close microphones, optional button, LED	Could add audio control but needs a complete device design	Compare companion against phone
Power and heat	Phone-only load must be measured	Low-power accessory target, <350 mW average	Multi-watt compute plus controlled power and thermal design	Do not build compute module in v1
Display and speaker	Phone screen and speaker	Same phone UI	Needs its own user interface or phone companion	Reuse phone
Hardware cost and certification	Lowest	Moderate, without battery or radio	Highest: SoM, memory, battery or external power, thermal design, enclosure	Companion only if it passes test
Product risk	Little physical differentiation	USB compatibility and audio route must be proven	Large hardware, supplier, thermal, and iOS risk	Android first

Two current embedded platforms illustrate the future Edge hardware class. Qualcomm documents the QCS6490 with up to 12 dense TOPS and up to 16 GB LPDDR5. MediaTek documents the Genio 1200 with 4.8 TOPS and support for up to 16 GB LP4X. These are examples, not selected parts or VoiceKey performance claims. A future Edge module proceeds only if it is faster or more consistent than the phone baseline and passes a 30-minute controlled power and heat test.

Compute platform	Published compute or memory context	Power and thermal boundary	VoiceKey role	Status
Google Pixel 8, Tensor G3, 8 GB RAM	Phone CPU, GPU, and supported Android runtime; no direct USB-to-NPU claim	Reuse phone battery and measure the full translation load	Reference Android compute platform for v1	Selected for benchmark
Qualcomm QCS6490	Up to 12 dense TOPS and up to 16 GB LPDDR5	Requires its own controlled power, RAM, board, and thermal design	Future Edge compute candidate	Research only
MediaTek Genio 1200	4.8 TOPS and up to 16 GB LP4X	Requires its own controlled power, RAM, board, and thermal design	Future Edge compute candidate	Research only

Android is first because its documented APIs cover USB host and accessory communication, USB audio, and microphone capture. A supported-phone matrix and route verification are still required. Android USB Android USB audio Android AudioRecord

iOS starts as a phone-only app using the built-in microphone and speaker through AVAudioSession. Apple's AccessoryAccess USB entitlement is for macOS USB access, not iOS accessories. A wired iPhone companion is outside v1 until a current Apple-supported iOS framework and hardware demonstrate it.

5.2 Form Factor, Components, and Power Budget

The v1 companion is a prototype, not a finished industrial product. It targets less than 20 g, about 45 x 22 x 9 mm, and a short flexible USB-C pigtail to reduce strain on the phone port. It has no display, battery, cellular radio, or model storage.

Component	Specification or class	Target active power	Role
USB-C attachment	Pigtail, passive USB-device CC path, ESD, USB 2.0 data	Low	Stable physical connection; no power pass-through in v1
MCU and audio bridge	Low-power USB-capable MCU	20 to 80 mW class	USB Audio Class, LED/button control, signed firmware update
Audio front end	Stereo voice codec or ADC class	20 to 60 mW class	Clocking, level management, and microphone interface
Close microphones	Two digital MEMS microphones	10 to 40 mW class	Test whether capture improves in the target setting
Button and LED	Optional button and visible status LED	<20 mW class	Mute or fallback control and recording state
Flash, PCB, passives, ESD	Firmware/configuration and protection	Low	No user audio storage and no model inference
Total companion	USB audio and interaction device	<350 mW average target	Measure on every supported phone before release

The limit is low because the phone supplies the power and AI compute. No all-day battery-life claim is made. Test three 30-minute active sessions on each reference phone: phone-only, companion attached but idle, and companion attached with active translation. Record battery-percent change, surface temperature, reconnects, actual audio route, and failures. Brownout, repeated USB enumeration, route loss, or unsafe heat fails the configuration.

5.3 Indicative BOM and Prototype Equipment

All figures are USD planning estimates based on public component listings. They are not supplier quotes, a final bill of materials, or a selling price.

The public anchors cover the STM32U575 MCU, TLV320AIC3263 codec, and SPH0655 digital microphone. The STUSB4500 is an alternative sink and PD-controller reference only. It is excluded from the v1 passive USB-device CC BOM unless the selected schematic explicitly changes to a controller-based topology. Assembly, enclosure, test, and manufacturing yield still need supplier quotes.

BOM block	Planning cost per unit	Boundary
USB connection	USD 1.50 to 4.00	One plug or pigtail, CC implementation, ESD, and connection-specific passives
MCU and audio bridge	USD 7.00 to 10.62	USB enumeration, reconnect, LED/button, and signed firmware control
Codec or audio front end	USD 7.38 to 13.58	Audio interface and signal-quality validation
Two digital MEMS microphones	USD 2.10 to 4.26	Compare with the phone microphone in target noise slices
Controls and PCB	USD 3.15 to 8.60	Button, LED, PCB, and non-USB passives
Enclosure, assembly, and test	USD 9.00 to 28.00	Shell, strain relief, low-volume assembly, functional test, and rework

The schematic must choose one Type-C configuration: a passive USB-device CC path or a named controller design. Never count both. Use the complete-unit bands in Section 3.3 for pilot planning until a configuration and two manufacturing quotes are selected.

Equipment	Why it is needed	Planning outlay
Three Android test phones	Compatibility, sustained inference, thermal, and audio-route matrix	USD 600 to 2,500 total
USB power meter and bench supply	Current, brownout, and phone-source measurement	USD 95 to 420
Logic analyzer and oscilloscope	USB timing, reset, rail, and signal debugging	USD 210 to 2,000+
Soldering and rework kit	EVT bring-up and board changes	USD 100 to 400
Basic audio and thermal checks	SPL or reference device plus IR check	USD 70 to 900
Prototype enclosure iteration	Fit, strain relief, and ergonomics	USD 50 to 300 per iteration

5.4 Hardware Release Gates

- Every supported phone passes tests for attachment, detachment, reconnection, screen lock, app restart, microphone-permission revocation, and Airplane Mode.
- The app confirms the actual capture route while recording. Requesting an external microphone is not enough. If the companion is not active, the app shows a phone-microphone fallback.
- A 30-minute session passes without brownout, route loss, repeated enumeration, crash, or unsafe thermal behavior.
- The companion provides a meaningful measured improvement over the phone-only baseline in the target noise condition. If it does not, VoiceKey remains phone-only.

6. System Architecture and Integration

6.1 Software Stack

Layer	Component	Role
Android OS	Android 13+ initial target	USB host, audio routing, microphone permission, display, storage, and thermal/battery signals
App	Kotlin and a native inference bridge	Conversation UI, language-pack setup, state machine, consent, and failure messages
Audio	Android audio stack and USB Audio Class companion	Phone microphone by default; companion only after actual route verification
Speech processing	Local VAD, ASR, translation, optional TTS	Run the selected English-Vietnamese stack offline after pack install
Pack management	Signed local pack registry	Version, hash, model and voice provenance, license record, and offline readiness
Privacy and security	Android Keystore and encrypted opt-in history	Local settings, deletion, and protection of retained history

Android capture follows documented microphone-permission and foreground-service requirements. If either is unavailable, VoiceKey stops capture and tells the user. [Android foreground-service requirements]

6.2 Architecture Diagram

Real-time translation flow

Phone or companion microphone

v

Speech detection and partial speech text

v

Final speech text after a natural pause

v

Local English-Vietnamese translation

v

Large bilingual transcript

v

Optional local TTS

The signed local language pack supplies speech, translation, and optional TTS resources.

The USB companion does not process the full AI pipeline in v1. The phone app manages the microphone state, language pack, model runtime, transcript, and all user-visible errors. This lets the team test translation without hardware and decide whether to keep the companion.

6.3 Offline-First Design Principles

- Each pack has a versioned manifest signed with a release key pinned in the app. Before activation, the app verifies the signature, file hashes, and revocation status. Key rotation and rollback are release tests.
- Runtime audio, speech recognition, translation, and optional TTS run without a network connection.
- A missing pack, invalid signature, unavailable microphone, or disconnected companion produces a clear local error, not a hidden cloud fallback.
- Audio is discarded after processing by default. History is an explicit user choice.
- Every pack records its version, hash, model and voice provenance, and release-license review.

7. Team Profile and Project Timeline

7.1 Required Team Roles

Before filing, replace these placeholders with real names, short qualifications, and contact details.

Team member	Role	Required experience	Contribution
To be named by submitting team	Product and field lead	User research, pilot design, operational safety	Scope, pilot success measures, consent, and partner coordination

Team member	Role	Required experience	Contribution
To be named by submitting team	Speech and translation lead	ASR, machine translation, evaluation, licensing	Model comparison, English-Vietnamese quality review, pack release ledger
To be named by submitting team	Android lead	Kotlin, mobile audio, local inference	Offline app, microphone/USB integration, phone matrix, privacy controls
To be named by submitting team	Hardware and firmware lead	USB-C, audio, PCB, embedded firmware	Companion EVT, power, LED/button, signed firmware, test fixture
To be named by submitting team	UX and test lead	Bilingual UX, usability test, accessibility	Transcript interface, failure states, study protocol, evidence demo

7.2 Project Timeline

Phase	Milestone	Key work	Target
1	Phone-only baseline	Choose reference phone, install local EN-VI pack, record offline baseline	Weeks 1 to 2
2	Model comparison	Compare ASR, translation, and optional TTS candidates; measure pack size, RAM, heat, and latency	Weeks 3 to 5
3	USB companion EVT	Build microphone, LED, and button prototype; verify attach, route, reconnect, and fallback	Weeks 6 to 8
4	Controlled demo	Run short two-person turns in Airplane Mode and in a network-enabled test that records zero VoiceKey outbound connections, with visible source text, translation, and timing	Weeks 9 to 10
5	Small supervised pilot	Test 10 users, then expand only if quality, heat, and companion benefit pass	Weeks 11 to 14
6	Decision report	Publish supported phones, results, failures, cost update, and Edge decision	Weeks 15 to 16

7.3 Demonstration Standard

The technical demo should be a real test, not a scripted marketing clip. Show the exact phone, Android build, app build, pack version, Airplane Mode, and actual microphone route. Also show source text, translated text, timing, one noisy turn, and one visible failure or fallback. Appendix C has a suggested shot list.

8. Submission Checklist

#	Checklist item	Current status
1	Executive summary, problem, users, and constraints are complete	Done

#	Checklist item	Current status
2	Business solution, comparison, price boundary, and pilot cost plan are complete	Done as a planning proposal
3	AI pipeline, candidate classes, optimization plan, robustness, and evaluation plan are complete	Done as a technical design; measurements pending
4	Hardware choice, indicative BOM, power target, and equipment needs are complete	Done as a technical design; EVT and supplier quotes pending
5	Architecture and offline-first rules are complete	Done
6	Team roster and contact details are complete	Pending user-provided names and credentials
7	Timeline is complete	Done as relative weeks; add calendar dates before filing
8	VoiceKey technical demo video is attached	Pending a real prototype recording
9	Final DOCX and PDF are exported and proofread	DOCX generated with this proposal; final PDF is pending team details and demo evidence

Appendix A. Research Sources and Claim Boundaries

Source	What it supports	What it does not prove
PhoST	English-Vietnamese speech translation benchmark and a cascade result in that study	VoiceKey accuracy, latency, or user value
PhoWhisper	Vietnamese ASR research and evaluation context	A commercial phone deployment
PhoMT	Vietnamese-English translation data and evaluation context	A shipping model license or product result
Whisper model card	Scope and runtime caveat for a baseline ASR family	Real-time behavior on a named VoiceKey phone
whisper.cpp and OPUS-MT EN-VI / VI-EN	Named evaluation candidates and local-runtime research starting points	Shipping rights, accuracy, and latency without the VoiceKey model comparison
sherpa-onnx	Local speech, VAD, Android, and TTS runtime research	Voice quality or commercial voice rights
Android USB and AOSP USB audio	Android USB and digital-audio mechanisms	Compatibility with every phone or accessory
Android AudioRecord	Microphone capture and routed-device verification research	A companion microphone being active merely because it was requested
Apple AVAudioSession	iOS phone-only microphone and speaker path	A custom wired iPhone accessory or generic USB-C accessory route

Source	What it supports	What it does not prove
QCS6490 and Genio 1200	Example embedded compute classes for a future Edge module	VoiceKey Edge performance or cost
NIST AI RMF	Human oversight framing for any future summaries or business drafts	Automated approval of an invoice, order, or safety action

Core academic references

- Nguyen et al. (2022), "A High-Quality and Large-Scale Dataset for English-Vietnamese Speech Translation", INTERSPEECH 2022. Supports the 508-hour EN-VI benchmark and its cascade-versus-end-to-end result.
- Le, Nguyen, and Nguyen (2024), "PhoWhisper: Automatic Speech Recognition for Vietnamese", ICLR 2024 Tiny Papers. Supports Vietnamese ASR candidate and evaluation context.
- Doan et al. (2021), "PhoMT: A High-Quality and Large-Scale Benchmark Dataset for Vietnamese-English Machine Translation", EMNLP 2021. Supports Vietnamese-English text-MT evaluation context.

Appendix B. Comparable Products and Public Demos

These links show related product formats and user flows. They are not VoiceKey benchmarks. A vendor tutorial can show a workflow, but not independent accuracy, latency, offline reliability, or microphone quality.

Platform	Product and direct link	What it is useful for	Evidence boundary
Product page	Timekettle W4 Pro	Phone-coupled earbud form factor and vendor pack workflow	Vendor product description only; not an EN-VI offline comparator or a benchmark
YouTube	Timekettle W4 Pro official tutorial	Setup and translation-mode flow	Official vendor workflow only
Product page and manual	Pocketalk S2 Plus and S2 setup manual	Screen-first handheld workflow and connectivity setup	Vendor documentation, not a benchmark
YouTube	Pocketalk How to Translate	Hands-on handheld translation steps	Official vendor workflow only
Product page	Vasco Translator E1 and Vasco Translator V4	Wearable and handheld product formats	Vendor product descriptions, not a benchmark

Platform	Product and direct link	What it is useful for	Evidence boundary
YouTube	Vasco Translator E1 official demo	Wearable conversation framing	Official vendor workflow only
Facebook	Timekettle W4 Pro vendor video	Secondary social workflow example	May be region or login restricted; vendor marketing only
Douyin	Timekettle W3 versus M3 official-store video	Vendor-described interaction difference: W3 hands-free speaking versus M3 button-triggered turns	Social store content; no accuracy, latency, or offline claim

Appendix C. VoiceKey Evidence Demo Shot List

Time	Shot	What it proves
0 to 10 seconds	Show the exact Android phone, app build, EN-VI pack version, and Airplane Mode. Include the recorded result of the network-enabled test that records zero VoiceKey outbound connections.	Test setup and offline precondition
10 to 20 seconds	Show the phone microphone route, then attach the companion and show the actual route change and LED state	The input route is verified, not assumed
20 to 40 seconds	English turn followed by Vietnamese turn; show partial text, final bilingual text, and an endpoint-to-result timer	Turn-based real-time flow under stated conditions
40 to 50 seconds	Play optional TTS and show that capture is paused	Echo-loop control in v1
50 to 65 seconds	Run one noisy turn and one missing-pack or disconnected-companion case	Honest failure and fallback behavior
65 to 80 seconds	Show device list, P50/P95 results, 30-minute session summary, and any failure count	Reproducibility boundary

Appendix D. Scope Limits and Release Conditions

- Do not describe turn-based translation as simultaneous, full-duplex interpretation until overlap, echo, heat, and latency tests pass.
- Do not claim a measured result, project demo, supplier quote, IP rating, certification, or commercial release before it exists.
- Do not claim a custom USB-C iPhone path. Validate phone-only iOS first.
- Do not ship a model or voice without a release ledger for weights, runtime, tokenizer, data obligations, and redistribution rights.

- Do not set a VoiceKey price from a raw BOM or a competitor retail page. Use selected hardware, two manufacturing quotes, warranty assumptions, and observed pilot support cost.
- Do not keep the USB companion because it is novel. Keep it only if the phone-only comparison shows a meaningful benefit.